

---- E-Sign Database Schema ----

1. TABLE SCHEMAS

Table 1: esign_providers: Stores provider configuration details used for E-Sign integration.

Column Name	Data Type & Constraints	Explanation
id	BIGSERIAL PRIMARY KEY	Unique provider identifier
provider_name	VARCHAR(100) UNIQUE NOT NULL	Provider name like Melento
provider_base_url	VARCHAR(500) NOT NULL	Base URL of provider
provider_endpoint	VARCHAR(200) NOT NULL	Provider API endpoint
encrypted_api_key	VARCHAR(500)	Encrypted provider API key
request_headers_json	JSONB	Dynamic request headers for provider
max_retry_attempts	INT DEFAULT 3	Maximum retry attempts before failover
priority	INT NOT NULL	Provider execution priority
is_active	BOOLEAN DEFAULT TRUE	Determines provider is active
webhook_supported	BOOLEAN DEFAULT TRUE	Indicates webhook availability
created_at	TIMESTAMPTZ DEFAULT NOW()	creation timestamp
updated_at	TIMESTAMPTZ	last updated timestamp

Table 2: esign_transactions: Stores final transaction outcome after E-Sign workflow execution completes.

Column Name	Data Type & Constraints	Explanation
id	BIGSERIAL PRIMARY KEY	Unique transaction identifier
provider_id	BIGINT NOT NULL FK	References From providers table
reference_id	VARCHAR(200) UNIQUE NOT NULL	Business transaction reference ID
docket_title	VARCHAR(255) NOT NULL	PDF Title
transaction_status	VARCHAR(50) NOT NULL	Status = Signed, Partially Signed, Expired, Failed
docket_id	VARCHAR(100)	From provider response
document_id	VARCHAR(100)	From provider response
expires_at	TIMESTAMPTZ	Signing link expiration timestamp
completed_at	TIMESTAMPTZ	Transaction completion timestamp
created_at	TIMESTAMPTZ DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMPTZ	Record last updated timestamp

Table 3: esign_signers: Stores signer details and signer-level signing information.

Column Name	Data Type & Constraints	Explanation
id	BIGSERIAL PRIMARY KEY	Unique signer identifier
transaction_id	BIGINT NOT NULL FK	References from esign_transactions table
signer_name	VARCHAR(255) NOT NULL	Signer full name
signer_email	VARCHAR(255)	Signer email address
signer_mobile	VARCHAR(20) NOT NULL	Signer mobile number
signer_status	VARCHAR(50) NOT NULL	Status = Signed , not signed
signed_at	TIMESTAMPTZ	Sign completion timestamp
page_number	INT NOT NULL	PDF page where signature appears
position_x	NUMERIC(10,2)	Signature X-axis position
position_y	NUMERIC(10,2)	Signature Y-axis position
Signer_ref_id	VARCHAR (200)	
Signer_id	VARCHAR (100)	
Invitation_link	TEXT	
created_at	TIMESTAMPTZ DEFAULT NOW()	Creation timestamp
updated_at	TIMESTAMPTZ	Updated timestamp

2. Folder Structure

```

ESign.API
|
|-- Controllers
|   |-- EsignController.cs
|
|-- Application
|   |-- Services
|   |   |-- Interfaces
|   |   |   |-- IEsignService.cs
|   |   |-- Implementation
|   |       |-- EsignService.cs
|   |-- DTOs
|   |   |-- Request
|   |   |   |-- PdfUploadRequest.cs
|   |   |   |-- EsignRequest.cs
|   |   |   |-- SignerRequest.cs
|   |   |   |-- WebhookRequest.cs
|   |   |-- Response
|   |       |-- PdfUploadResponse.cs
|   |       |-- EsignResponse.cs

```

```
└─ WebhookResponse.cs

└─ Domain
  └─ Entities
    ├── Provider.cs
    ├── EsignTransaction.cs
    └─ EsignSigner.cs

└─ Infrastructure
  ├── Cache
  │   └─ ProviderCacheService.cs
  ├── Repositories
  │   ├── Interfaces
  │   │   ├── IProviderRepository.cs
  │   │   ├── IEsignTransactionRepository.cs
  │   │   └─ IEsignSignerRepository.cs
  │   └─ Implementation
  │       ├── ProviderRepository.cs
  │       ├── EsignTransactionRepository.cs
  │       └─ EsignSignerRepository.cs
  └─ Dapper
      └─ DapperContext.cs

└─ Middleware
  ├── CorrelationMiddleware.cs
  ├── RequestResponseLoggingMiddleware.cs
  └─ ExceptionMiddleware.cs

└─ Utilities
  ├── AppLoggers.cs
  ├── EncryptionHelper.cs
  ├── LogPathHelper.cs
  ├── MaskingHelper.cs
  ├── PdfHelper.cs
  ├── SerilogFactory.cs
  └─ ValidationHelper.cs

└─ Constants
  └─ StatusConstants.cs

└─ appsettings.json
└─ appsettings.Development.json
└─ Program.cs
```